

◆ AWS CLOUD PRACTITIONER BOOTCAMP

THURSDAY — SESSION 2 LABS

Shared Responsibility + Risk Awareness

[Student Hands-On Console Labs](#)

◆ LAB OVERVIEW

These labs are designed to help students understand:

- ✓ Shared Responsibility Model
 - ✓ IAM basics
 - ✓ S3 access control risks
 - ✓ Security Group exposure risks
 - ✓ Governance awareness
 - ✓ Cloud misconfiguration concepts
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◆ IMPORTANT LAB RULES

Before starting:

- ✓ Use your AWS Free Tier account
 - ✓ Do NOT upload personal or sensitive data
 - ✓ Delete resources after completing labs
 - ✓ Follow instructions carefully
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◆ LAB 1 — IAM USER + MFA SETUP

Objective

Understand identity management and access control.

◆ WHAT YOU WILL LEARN

- ✓ Why IAM matters
- ✓ Difference between root account and IAM user
- ✓ Basic least-privilege concepts
- ✓ MFA protection

◆ STEP 1 — OPEN IAM

In AWS Console:

Search for:

IAM

Open:

[Identity and Access Management \(IAM\)](#)

◆ STEP 2 — CREATE IAM USER

Navigate to:

Users → Create User

[Configuration](#)

Username

Create:

`student-user`

[Access Type](#)

Enable:

☒ AWS Management Console access

[Password](#)

Choose:

☒ Custom password

Important

Disable:

✓ User must create a new password at next sign-in
(optional for lab simplicity)

◆ STEP 3 — ADD PERMISSIONS

Choose:

Attach policies directly

Select:

AmazonS3ReadOnlyAccess

◆ STEP 4 — REVIEW & CREATE

Click:

Create User

◆ STEP 5 — ENABLE MFA

Open the new IAM user.

Navigate to:

Security Credentials → Assign MFA Device

Choose:

✓ Authenticator App

Recommended:

- Google Authenticator
 - Microsoft Authenticator
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◆ STUDENT OBSERVATION

Notice:

- IAM controls who can access AWS
 - Permissions determine what users can do
 - MFA reduces unauthorized access risk
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◆ LAB 1 DISCUSSION QUESTIONS

1. Why should the root account not be used daily?
 2. What risk exists without MFA?
 3. Why is least privilege important?
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◆ LAB 2 — S3 PUBLIC ACCESS RISK

Objective

Understand cloud storage permissions and exposure risk.

◆ WHAT YOU WILL LEARN

- ✓ S3 bucket creation
 - ✓ Public access settings
 - ✓ Why public exposure is dangerous
 - ✓ Governance awareness
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◆ STEP 1 — OPEN S3

Search:

S3

Open:

[Amazon S3](#)

◆ STEP 2 — CREATE BUCKET

Click:

Create bucket

Bucket Name

Example:

`yourname-risk-lab`

Bucket names must be globally unique.

Region

Leave default.

◆ STEP 3 — BLOCK PUBLIC ACCESS

You will see:

✓ Block all public access

IMPORTANT:

Keep this ENABLED initially.

Create bucket.

◆ STEP 4 — UPLOAD FILE

Upload:

- Simple text file
- OR
- Practice document

Do NOT upload sensitive information.

◆ STEP 5 — EXPLORE PERMISSIONS

Open:

Permissions tab

Review:

- Bucket Policy
 - Public Access Settings
 - Access Control Lists (ACLs)
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◆ OPTIONAL DEMO (INSTRUCTOR GUIDED)

Temporarily disable:

✓ Block Public Access

Observe:

- AWS warning messages
- Public exposure indicators

Re-enable protection immediately after observation.

◆ STUDENT OBSERVATION

Understand:

- AWS does not automatically decide your access policy
 - Misconfiguration can expose data publicly
 - Customers are responsible for access control
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◆ LAB 2 DISCUSSION QUESTIONS

1. Why are public buckets dangerous?
 2. Did AWS stop you from making the bucket public?
 3. Who is responsible for securing bucket access?
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◆ LAB 3 — SECURITY GROUP RISK AWARENESS

Objective

Understand network exposure risk.

◆ WHAT YOU WILL LEARN

- ✓ Basic EC2 deployment
- ✓ Security Group concepts
- ✓ Inbound traffic rules
- ✓ Internet exposure risk

◆ IMPORTANT NOTE

This lab demonstrates exposure concepts.
Do NOT leave risky configurations active after completion.

◆ STEP 1 — OPEN EC2

Search:

EC2

Open:

[Amazon EC2](#)

◆ STEP 2 — LAUNCH INSTANCE

Click:

Launch Instance

[Configuration](#)

Name

risk-awareness-lab

[AMI](#)

Choose:

- ✓ Amazon Linux
-

Instance Type

Choose:

✓ t2.micro (Free Tier eligible)

◆ STEP 3 — CREATE SECURITY GROUP

Create new Security Group.

Add Rules

Allow SSH

Type:

SSH

Port:

22

Source:

0.0.0.0/0

◆ IMPORTANT DISCUSSION

This means:

Anyone on the internet can attempt connection.

This is usually NOT recommended for production environments.

◆ STEP 4 — REVIEW WARNINGS

AWS may display warnings regarding open access.

Read and understand them carefully.

◆ STEP 5 — LAUNCH INSTANCE

Launch EC2 instance.

Observe:

- Public IP
 - Security Group rules
 - Internet accessibility
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◆ STEP 6 — FIX THE RISK

Edit inbound rule.

Replace:

0.0.0.0/0

With:

My IP

◆ STUDENT OBSERVATION

Understand:

- Security Groups act as virtual firewalls
 - Open ports increase attack surface
 - Governance helps standardize safe configurations
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◆ LAB 3 DISCUSSION QUESTIONS

1. Why is 0.0.0.0/0 risky?
 2. What is an attack surface?
 3. How does governance reduce this risk?
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◆ LAB 4 — ROOT ACCOUNT SECURITY REVIEW

Objective

Understand root account risks and security best practices.

◆ STEP 1 — OPEN ACCOUNT MENU

Top-right account menu →

Security Credentials

◆ STEP 2 — REVIEW ROOT ACCOUNT STATUS

Check:

- ✓ MFA enabled?
 - ✓ Access keys active?
 - ✓ Strong password used?
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◆ STEP 3 — UNDERSTAND ROOT ACCOUNT RISK

The root account has:

- Full AWS access
- Unlimited permissions
- Billing control

If compromised:

Entire environment is compromised.

◆ BEST PRACTICE

- ✓ Use IAM users daily
 - ✓ Protect root account with MFA
 - ✓ Avoid root access keys
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◆ LAB 5 — GOVERNANCE OBSERVATION EXERCISE

Objective

Develop governance thinking.

◆ STUDENT TASK

Review all previous labs and identify:

Resource	Risk	Governance Control
IAM User	Excessive permissions	Least privilege
S3 Bucket	Public exposure	Block public access
Security Group	Open SSH	Restricted IP access

◆ FINAL STUDENT CHALLENGE

Answer:

1. What is the Shared Responsibility Model?
 2. What does AWS secure?
 3. What must customers secure?
 4. Why are misconfigurations dangerous?
 5. How does governance reduce cloud risk?
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◆ CLEANUP INSTRUCTIONS (IMPORTANT)

Students must delete resources after labs.

Delete:

- ✓ EC2 instances
 - ✓ Security Groups (if unused)
 - ✓ S3 buckets and uploaded files
 - ✓ IAM test users (optional)
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◆ FINAL PRINCIPLE

AWS provides secure infrastructure.

Customers must configure and govern that infrastructure responsibly.